

YADAV S V

EMBEDDED FIRMWARE ENGINEER



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Bangalore

Summery

Results-oriented Embedded Firmware Engineer with more than 2 years experience in design, development, testing, and implementation of various embedded software systems. I am keen to learn and implement designing and building applications with usability and high performance in mind.

Skills

- **Programming Languages:** C, Embedded-C, Python, Verilog and Perl
- **Embedded skills** LL Driver Development of GPIO, SPI, I2C and UART. Fast prototype Development using MBED and Arduino.
- **Testing and Debugging** CANAnalyser, ST-Link, Segger Jlink.
- **Version Control Tool** GIT, Helix ALM, Empower QLM.
- **Software's and OS's:** STMCubeIDE, Arduino IDE, MP Lab, Eclipse based IDE, Altium Designer, MATLAB, Xilinx, Windows, Linux (Ubuntu).
- **MCU** NVIDIA Jetson nano, STM32F Series, STM32L series, ESP32, Arduino

Experience

RGBSI Pvt Ltd. (April 2021 - Till date)

Engineer

1. Door Control Unit (DCU) :

A MCU-based Door Control Unit for locomotive applications That interfaces with the Train Control and Management System (TCMS) to communicate with the doors of every coach and is automated with opening and closing actions and additional safety features. and also be certified with SIL-2 of the EN50128 standard.

Contribution :

- Developed an embedded C API for USB MSC using STM32CubeIDE and ST Link debugger.
- Responsible for maintaining CM on Helix ALM, the version control tool.
- Involved in a complete SDLC (V-Cycle)
- Gone through the complete EN50128 standard document to make the checklists required for getting certified under SIL level 2.
- Reviewing and understanding the customer requirements
- Reviewing the software requirements

2. ADAS and DMS :

The Assistant Systems for drivers have several functionalities with alert warnings and driver management systems with smart dashboards to view the performance of a driver based on their driving performance.

Contribution :

- Annotated data-set to feed and train the system.
- Developed a gstreamer pipeline application for enhancing the performance of the camera interface on the NVIDIA Jetson nano.

3. CAN Diagnostics ISO 14229 :

Black-BOX testing and diagnostics based on ISO 14229 for the ECU designed by Ford Motors using the CANAnalyser tool and Ford Motor 's proprietary tool.

Contribution :

- Perform testing on the ECU based on the test cases provided by the test lead by using Vector CANoe.
- Generating the report by CANAnalyser and Ford's proprietary tool.
- reporting the bug whenever necessary and communicating with the client.
- Automated some repetitive test cases by using the Python AutoGUI library.

4. Row Display Unit :

Reverse Engineering and Developing a Row Display Unit using a Microchip 32-bit PIC controller.

Contribution :

- Build a quick test setup for RDU.
- Performed various hardware and software tests to find the root causes and fix the bug.
- Coordinated with the hardware team.

5. Intelligent Transportation System:

Developing an SCU for an advanced application for providing intelligence for transport enables users to be better informed and make safer, more coordinated, and "smarter" use of transport networks.

Contribution :

- Understanding the UDS-11 Specification
- Made a checklist for the requirements specified in the standard document.
- Performed testing and reporting the bug.
- worked with the mechanical team to build a prototype for client demonstration.

HONEY WORKS LLP (June 2020 - March 2021)

Production Executive Trainee

1. Automatic Faucet :

IR-based Automatic Faucet with Solenoid Valve for turning ON and OFF water flow using the ATmega328P (IR transmitter as IR LED and IR receiver as IR Photo Transistor) and a PCB designed and developed as a product.

Contribution :

- Performing market research about the product.
- Developing market requirement of the product.
- Discussing with top management and developing prototypes.
- Designing and developing prototype for the displaying the product functionality.
- Designing a 2- Layer PCB for the product.
- Developed the code on ATmega328p.

2. Smart Touch sensor based Mirror

For consumer electronics applications, a touch sensor-based mirror with LED and Defogger for the powder room. That enables LED's light and defogger on a single touch. and with various modes of operation like dim mode, bright mode.

Contribution :

- Research about the similar product.
- Developing the product for production release
- Planning for production release and assembly area.
- Contacting vendors for the components and modules.

Education

Degree	Major	College and University	Score	Year
B.E	E & C Engineering	MS Engineering College VTU University	7.45cgpa	2020
PUC	PCME	Presidency PU College Karnataka PU Board	77%	2016
SSLC		Karnataka State Board	84.32%	2014

Achievements and Activities

- Awarded as **Budding Young engineer** at the college in the year 2017.
 - Conducted several Inter and Intra College workshops on PCB Design, Cymplex VLSI Tool & Basics of FPGA design.
 - Participated in **IEEE YESIST12 Project Innovation Challenge** at VidyaVikas Institute of Engineering and Technology Mysore,
 - Participated in 4-day National Seminar On "Innovations in Science and Engineering" at Sir. M. Visvesvaraya Institute of Technology and Inter College Robot and Drone Exhibition.
 - Participated in 3 days National Science and Technology Entrepreneurship Development Board (NSTEDB), Dept. of Science and Technology, Govt. India.
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Academic Projects

Smart system for early detection of fish kill and prevention

- A IoT based smart system for preventing the fish kills based on the external changes that takes place in the environment (such as increasing in the temperature, decreasing the oxygen level of water and increasing in the algae and fallen leaves that causes decaying activity in the lakes etc.) that is harmful for the survival of fishes. Which is difficult for the cultivation of fishes. Our idea of building a complete IoT based system and observe the data that gather on the clouds by the sensors such as DO, Temperature and fish-count. Implemented MQTT protocol and involved in designing smart system with several sensor and actuators. Language used- Embedded C

VLSI Design of LOW power High-Speed FIR Filter

- A project on designing Low Power high speed FIR Filter with 0.35um open-source standard cell set by making using of FDA Matlab tool, then simulating the obtained code on Xilinx later optimizing the code on Qflow (open-source tool) and recorded the Area, Timing analysis of different combination coefficients of filter. Programming Language used- Verilog

Designed a PCB for Water level sensor

- PCB design for Water Level sensor that has series of ten exposed copper traces having power and sense aisle between each other rails the value varies as the resistance varies by increasing or decreasing of water level by the law of conductivity of water.

DECLARATION

I hereby declare that the information furnished above is true and authentic to the best of my knowledge and belief.